

Arithmetic & Algebra

Basic identities, quadratic formula, etc.

Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial Theorem:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Arithmetic Sequence:

$$a_n = a_1 + (n - 1)d$$

Geometric Sequence:

$$a_n = a_1 \cdot r^{(n - 1)}$$

Geometry & Trigonometry

Area formulas, Pythagorean theorem, sine/cosine laws.

Pythagorean Theorem:

$$a^2 + b^2 = c^2$$

Area of a Circle:

$$A = \pi r^2$$

Sine Rule:

$$a/\sin(A) = b/\sin(B) = c/\sin(C)$$

Cosine Rule:

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$$

Calculus

Derivatives, integrals, fundamental theorems.

Derivative Definition:

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

Integral Definition:

$$\int f(x) dx = F(x) + C$$

Fundamental Theorem of Calculus:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Chain Rule:

$$d/dx [f(g(x))] = f'(g(x)) * g'(x)$$

Linear Algebra

Matrix operations, eigenvalues, vector spaces.

Matrix Multiplication:

$$(AB)_{ij} = \sum_k A_{ik} * B_{kj}$$

Determinant of a 2x2 Matrix:

$$\det(A) = ad - bc$$

Eigenvalue Equation:

$$A * v = \lambda * v$$

Dot Product:

$$a \cdot b = \sum_i a_i * b_i$$

Probability & Statistics

Bayes' theorem, normal distribution, expected value.

Bayes' Theorem:

$$P(A|B) = [P(B|A) * P(A)] / P(B)$$

Normal Distribution:

$$f(x) = (1 / \sqrt{2\pi}) * e^{-(x - \mu)^2 / (2\sigma^2)}$$

Expected Value:

$$E(X) = \sum [x * P(x)]$$

Variance:

$$\text{Var}(X) = E[(X - \mu)^2]$$

Number Theory

Modular arithmetic, prime theorems.

Fermat's Little Theorem:

$$a^{(p-1)} \cdot 1 \pmod{p}$$

Euler's Totient Function:

$$\cdot(n) = n * \cdot (1 - 1/p)$$

Prime Number Theorem:

$$\cdot(x) \sim x / \log(x)$$

Chinese Remainder Theorem:

$$x \cdot a_i \pmod{n_i}$$

Differential Equations

First/second-order ODEs, PDEs.

First-Order ODE:

$$dy/dx + P(x)y = Q(x)$$

Second-Order ODE:

$$d^2y/dx^2 + P(x)dy/dx + Q(x)y = R(x)$$

Wave Equation:

$$\partial^2 u / \partial t^2 = c^2 \partial^2 u$$

Heat Equation:

$$\partial u / \partial t = \partial^2 u$$

Abstract Algebra & Topology

Group axioms, homomorphisms.

Group Axioms:

$G = (G, *)$ with closure, associativity, identity, and inverses

Ring Axioms:

$R = (R, +, *)$ with distributivity, associativity, and identity

Homomorphism:

$f: G \rightarrow H$ with $f(x * y) = f(x) * f(y)$

Euler Characteristic:

$$\chi = V - E + F$$

Applied Mathematics

Navier-Stokes, Schrödinger equation, Black-Scholes.

Navier-Stokes Equation:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

Schrödinger Equation:

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi$$

Black-Scholes Equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Fourier Transform:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$